
2019-Ph-H1-Q15

Section: Our Dynamic Universe

Topic: Energy and Power

Question:

A car engine does 6000 J of work in 3 seconds. What is its power output?

Answer:

$$P = \frac{W}{t} = \frac{6000}{3} = 2000 \text{ W}$$

Correct Option: E (2000 W)

Guidance for Students:

Work done ÷ time = power. Be sure energy is in joules and time in seconds.

Revision Tips:

- 1 watt = 1 joule per second
 - Double-check whether time is in minutes or seconds
 - Power is a **rate** of doing work
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